QMEA

1.Which of the following best describes a Binomial Distribution?

A. A distribution used to describe continuous measurements

B. A distribution used to count the number of successes in a fixed number of independent trials

C. A distribution used to count the number of events occurring within a fixed interval of time or space

D. A distribution used only when the sample size is large

2. Which of the following best describes a Poisson Distribution?

A. A distribution used to count the number of successes in a fixed number of trials

B. A distribution used to describe continuous measurements

C. A distribution used to count the number of events occurring within a fixed interval of time, area, or space

D. A distribution used when the population mean is unknown

3. Which of the following best describes a Normal Distribution?

A. A continuous probability distribution that is symmetric and bell-shaped around its mean

B. A distribution used to count rare events over time

C. A distribution used only for binary outcomes

D. A distribution in which all values have the same probability

4. A treatment has a 70% probability of success. If the treatment is given to 20 patients, which distribution should be used to describe the number of patients who respond successfully?

A. Binomial

B. Poisson

C. Normal

5. An emergency department receives an average of 5 patients per hour. Which distribution should be used to describe the number of patients arriving during the next hour?

A. Binomial
B. Poisson
C. Normal

6. The systolic blood pressure of adults has a mean of 120 mmHg and a standard deviation of 15 mmHg and is approximately bell-shaped. Which distribution is most appropriate?

A. Binomial

B. Poisson

C. Normal

7. Each surgical patient has a 5% probability of developing a postoperative infection. Among 50 patients, which distribution should be used to describe the number who develop an infection?

A. Binomial
B. Poisson
C. Normal

8. A hospital experiences an average of 2 cardiac arrests per week. Which distribution should be used to describe the number of cardiac arrests occurring next week?

A. Binomial
B. Poisson
C. Normal